How Much Does Hearing the Future Buy? Ultra-Low-Latency Causal Binaural Speech Enhancement

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Abstract

Hearing aids and hearables must enhance speech within about 5 ms, so they cannot use the future context that strong separators rely on. We ask how much a strictly causal binaural enhancer loses as its algorithmic latency drops from 32 to 1 ms, and where that loss comes from. Real speech and noise are spatialised through measured heads in simulated rooms. We compare oracle masks, causal networks (an asymmetric-window GRU with deep filtering over past frames, a symmetric-window twin, and a binaural Conv-TasNet), their non-causal twins, and hop-matched lookahead. The window costs little: an ideal mask gains 6.88 dB SI-SDRi at 1 ms against 7.88 dB at 32 ms. The future costs almost nothing: at 4 ms, non-causal and causal twins differ by 0.03 dB, and matched-hop lookahead changes quality by under 0.05 dB. Learning dominates: trained models remain 3.64 dB below the oracle. Interaural level differences stay within the 1 dB JND, but no model, and not even the oracle, keeps the ITD error under 20 μs (best 47.8 μs). A 4 ms causal model streams at real-time factor 0.21 on one CPU thread. Short latencies are not what holds small causal enhancers back; training and binaural timing fidelity are.

1 Introduction

The neural separators that define the state of the art, such as Demucs [3], Wave-U-Net [17] and non-causal Conv-TasNet [11], decide what a sample is from a context that reaches into the future. A hearing aid or a hearable cannot wait. Its output is heard together with the direct sound leaking past the earpiece, and once the device delay grows beyond a few milliseconds the listener hears comb filtering, echo, or a doubled own voice. Tolerance studies put the usable budget at roughly 5–10 ms in total, and lower for open fittings [18, 19]. Such a device must therefore be *strictly causal*, with a few milliseconds of buffering at most. If it is binaural, it must also keep the interaural time and level differences (ITD, ILD) that let the listener place the talker and unmask speech in noise [4].

This paper prices that constraint. On real speech and noise spatialised through measured heads in simulated rooms, we sweep the *algorithmic latency* L (synthesis window plus lookahead) from 1 to 32 ms and measure, at each L: (i) how good *any* mask can be (an oracle ratio mask, with symmetric and asymmetric windows); (ii) what causal networks reach: an asymmetric-window GRU with deep filtering over past frames (AsymDF), the same network with a short symmetric window, and a causal binaural Conv-TasNet; (iii) what the same networks gain when allowed to hear the whole utterance; (iv) whether a fixed budget is better spent on window length or on lookahead; and (v) the measured CPU cost of streaming at batch size one.

Our hypothesis was that below about 8 ms the loss is dominated by frequency resolution rather than by the missing future. If so, spending capacity on the *past* (a long analysis window, recurrent state, filtering over past frames) should recover most of it, keeping the 4 ms causal model within 1 dB SI-SDRi of the 32 ms one and its target cues within the perceptual just-noticeable differences (JNDs). Section 4 reports which parts of this survive.

2 Related work

Low-latency enhancement. Conv-TasNet [11] replaced the STFT with a short learned filterbank, which made millisecond-scale latency possible in a time-domain separator. Its causal variant reports a single operating point (2 ms at 8 kHz). STFT-domain systems reach low latency with asymmetric analysis and synthesis windows [12]. Wang et al. [22] showed that a DNN with a long asymmetric analysis window and a 2–4 ms synthesis window keeps most of the quality of a 32 ms system. DeepFilterNet [16] adds complex filtering over neighbouring frames, which we restrict to *past* frames.

Binaural enhancement and cue preservation. Classical binaural noise reduction keeps cues by applying one real gain to both ears, or by constraining the binaural multichannel Wiener filter [4]. Han et al. [7] preserved

spatial cues in a real-time binaural Conv-TasNet, and Tokala et al. [21] trained with explicit ILD/IPD losses. Both report one latency.

What is new here. A single controlled sweep that separates three costs:

- the window (an oracle bound),
- the future (causal vs. non-causal twins), and
- learning (oracle vs. non-causal model).

The sweep also includes a fixed-budget window-vs-lookahead split, target and residual-noise cue errors against JNDs, and measured single-thread real-time factors with streaming outputs verified to equal the offline ones.

3 Methods

3.1 Causal framing and latency

Every system shares one framing. With hop R and analysis length N, frame t covers input samples $[(t{+}1)R{-}N,\ (t{+}1)R)$, so it ends at the newest sample. Only the last $2R$ samples of each frame are synthesised and overlap-added at hop R. The product of analysis and synthesis windows there is a periodic Hann of length $2R$, which gives perfect reconstruction. An output sample is final once the next frame has been processed, so

$$L = 2R + DR, \tag{1}$$

with D frames of lookahead. Compute time is reported separately as a real-time factor (RTF). The asymmetric analysis window [12, 22] rises over $N - R$ samples ($N = 32$ ms) and falls over R. With $N = 2R$ it reduces to the ordinary sqrt-Hann pair. For every learned model, we test that streaming chunk by chunk reproduces the offline output exactly after a delay of $L - R$, and that changing future input leaves earlier outputs unchanged.

3.2 Systems

- **AsymDF**: per-frame compressed complex spectra of both ears ($|X|^{0.3}e^{j\angle X}$), frame-wise LayerNorm, a linear layer, two unidirectional GRU layers (128 units), and a per-ear complex deep filter [16] over the current and two past frames.
- **Sym. window DF**: the same network with $N = 2R$.
- **Conv-TasNet** [11]: a binaural causal variant with a shared learned basis of length $2R$ (at least $4R$ filters), a 2×6-block dilated TCN with per-frame LayerNorm (bounded streaming state), and per-ear masks.
- **Non-causal twins**: bidirectional GRUs and centred convolutions.
- **Common-gain head**: one real gain applied to both ears, which preserves cues by construction.
- **Oracle**: the ideal ratio mask $\sqrt{|S|^2/(|S|^2 + |X{-}S|^2)}$, per ear or common, under each window.
- **Binaural Wiener**: decision-directed [5] common gain with minimum-tracking noise estimation.

Training. The loss is the joint binaural SNR plus energy-weighted $|\Delta\text{ILD}|$ and $1 - \cos\Delta\text{IPD}$ (below 1.5 kHz). The SNR depends on scale, so neither ear can change the ILD for free. Optimisation uses AdamW at 10^{-3} with a cosine decay and 100 warm-up steps. TasNet starts from a pass-through basis with a 10× lower basis learning rate, so every family starts at the identity.

3.3 Data

Speech is LibriSpeech [14] (4.0 h for training, speaker-disjoint splits), and noise is DEMAND [20]. We render binaural room impulse responses by passing every image source [2] of a pyroomacoustics shoebox [15] (order 6, RT60 0.2–0.8 s) through the measured KEMAR [6] HRIR of its arrival direction. A level-matched binaural diffuse tail decays at the room's RT60. There are 150 training rooms, each with one talker (0.5–2.5 m, any azimuth) and six noise directions. The **target** is the direct path plus reflections within 50 ms. Mixtures are drawn fresh on the GPU at SNR −5 to 15 dB.

- **Test set**: 300 × 3 s, with unseen speakers and rooms.
- **Held-out set**: 150 mixtures with a different head (CIPIC [1]), unseen rooms, and unseen noise environments.

3.4 Metrics

- **Quality**: SI-SDR improvement [10] and ESTOI [8], both averaged over the ears.
- **Target cue errors**: $|\Delta\text{ITD}|$ from band-limited (0.1–1.5 kHz) GCC-PHAT with a 1/32-sample peak search, and broadband $|\Delta\text{ILD}|$, both against the clean target. The JNDs are 20 μs and 1 dB [9, 13].
- **Statistics**: means with bootstrap 95% CIs over utterances, and paired Wilcoxon tests between adjacent latencies.
- **Compute**: the median and p99 time per chunk while streaming on one CPU thread at batch size one.

4 Results

All numbers are means over the 300 test mixtures with bootstrap 95% CIs, unless marked *held-out* (150 mixtures, unseen head and noise). Table 1 lists every cell.

4.1 What the window costs: the oracle bound

With an asymmetric window, the ideal ratio mask gains 6.88 dB SI-SDRi at 1 ms and 7.88 dB at 32 ms (Fig. 1). Going from 32 ms down to 1 ms therefore costs about 1 dB even for a perfect mask. A symmetric window loses more (6.15 dB at 1 ms), and the asymmetric analysis recovers 0.73 dB of it at 1 ms and 0.41 dB at 4 ms. The effect is clearer in intelligibility: oracle ESTOI at 1 ms is 0.851 with the asymmetric window against 0.781 with the symmetric one. Frequency resolution is a real cost, but a small and mostly recoverable one.

4.2 What the future buys: almost nothing

At 4 ms, the non-causal AsymDF, which sees the whole utterance, scores 2.55 dB against 2.54 dB for its causal twin. After 12k steps the scores are 3.60 vs. 3.57 dB (Fig. 2). Conv-TasNet shows the same at 4 ms (2.20 vs. 2.10 dB) and gains at most a few tenths of a dB at 32 ms (1.96 vs. 1.71 dB). Explicit lookahead tells the same story (Fig. 4). At a fixed 8 ms budget, spending latency on lookahead looks better (2.15 dB with D=0 vs. 2.73 dB with D=6), but those runs also have smaller hops. At a *matched* hop, lookahead changes SI-SDRi by 0.03 dB (+4 ms of future, 95% CI [0.01, 0.06]) and -0.03 dB (+6 ms, CI [-0.06, -0.01]). The future is not what limits these models.

4.3 What learning costs: everything else

The gap that remains is learning. At 4 ms the oracle exceeds the non-causal model by 4.69 dB after 2k steps and by 3.64 dB after 12k (Fig. 2), against 0.64 dB for the window and 0.03 dB for the future. The asymmetric window's oracle advantage does not show up in the trained models either: the symmetric twin is within 0.2 dB at every latency (2.73 vs. 2.76 dB at 2 ms). At this model size and training budget, a 4 ms device would gain more from a better-trained network than from any amount of latency.

Shorter latency trains faster. At equal steps, quality *falls* with latency in every family (paired Wilcoxon, AsymDF 2→4 ms Δ=-0.22 dB, p=4.6e-25; 8→32 ms Δ=-1.06 dB, p=1.6e-48). This is the opposite of the oracle trend. A 1 ms hop gives 16× more supervised frames per step than a 16 ms hop, so equal steps are not equal training. Six times more steps did not close it either: the 32 ms model reaches 1.98 dB and the 4 ms model 3.57 dB. The pre-registered gap criterion (4 ms within 1 dB of 32 ms) is therefore met (gap -1.59 dB, 12k-step budget), but for the wrong reason, and we do not count it as support.

4.4 Interaural cues

Level cues survive (Fig. 3). Every causal model keeps the target |ΔILD| below the 1 dB JND, down from 1.21 dB in the mixture (0.36 dB for the 12k-step 4 ms AsymDF). Timing cues do not. No trained model gets |ΔITD| near 20 μs: the best is 47.8 μs, against 73.8 μs unprocessed. Even the oracle sits at 21.7 μs, while the estimator floor is only 3.8 μs, so the oracle's error is real. Late reverberation and noise left in low-frequency bins pull the broadband ITD. The common-gain head does not fix this on the training head (63.4 μs), but it generalises best to the unseen head: held-out SI-SDRi is 1.58 dB against 0.44 dB for the per-ear deep filter.

4.5 Generalisation

Every model loses most of its gain on the held-out set, for example 0.44 dB at 4 ms (AsymDF) and 1.10 dB (Conv-TasNet), against 2.54 and 2.10 dB on test. The per-ear deep filter on the long asymmetric window transfers worst. The symmetric-window filter (0.79 dB at 4 ms), Conv-TasNet and the common gain keep more. Fine per-ear spectral detail is what is most tied to one head's HRTF, so a new head and new noise expose it first.

4.6 Real-time cost

Streaming AsymDF takes 0.423 ms per chunk regardless of hop (Fig. 5), so it keeps up down to 1 ms latency (RTF 0.85 median, 0.99 p99) on one thread of AMD64 Family 25 Model 33 Stepping 2, AuthenticAMD. At 4 ms it runs at RTF 0.21, so the delay heard is about 4 + 0.423 ms. The constant per-chunk cost shows the time is dominated by per-call framework overhead, not by arithmetic (371 MMAC/s at 4 ms). Conv-TasNet's 12 streamed TCN blocks cost 2.457 ms per chunk, so it is real-time only from 8 ms upward (RTF 1.23 at 4 ms, 0.61 at 8 ms). A fused or compiled kernel would change this constant, not the algorithmic latency.

5 Discussion

Verdict on the hypothesis. We expected the low-latency loss to come mainly from frequency resolution rather than from the missing future. The decomposition supports half of that. The future is essentially worthless to these models: non-causal twins and matched-hop lookahead change SI-SDRi by a few hundredths of a dB. But the window is not the main cost either. The oracle says a 4 ms window costs 0.64 dB, while learning costs 3.64 dB even after 12k steps. The 1 dB gap criterion is

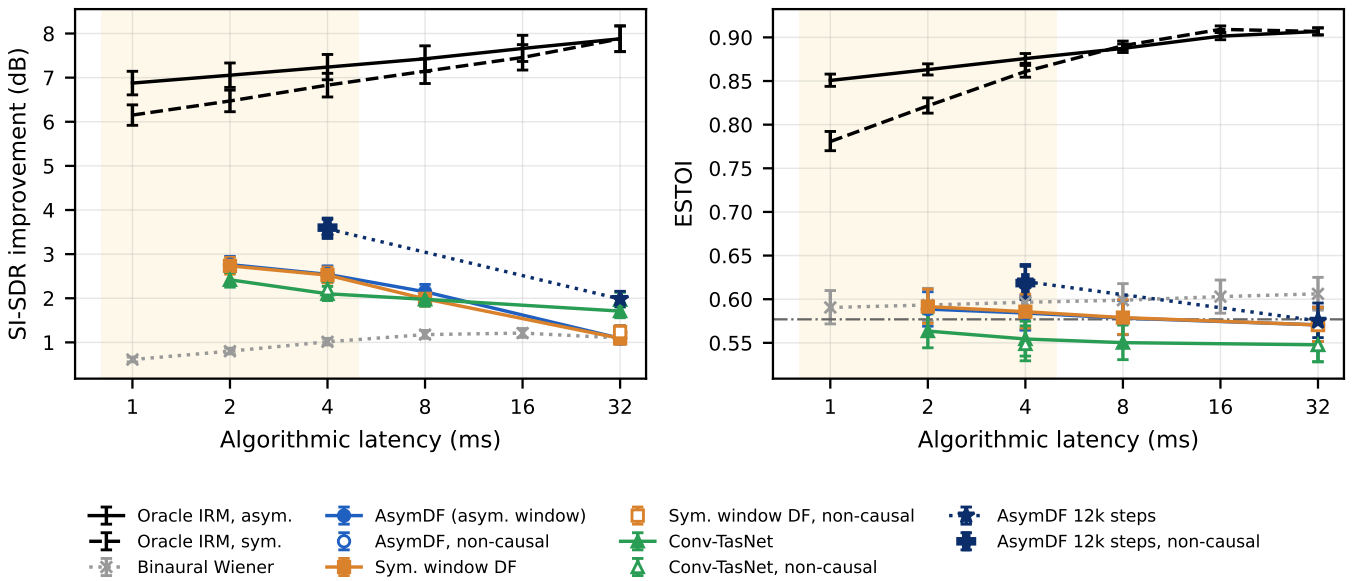


Figure 1: Quality against algorithmic latency. The shaded band is the hearing-aid budget ($\leq$5 ms). Solid markers are causal models (2k steps); hollow markers are non-causal twins; stars are the 12k-step AsymDF runs. The symmetric-window family coincides with AsymDF at 32 ms, where the two windows are identical. Black lines are oracle ratio masks.

Model	L (ms)	SI-SDRi (dB)	ESTOI	|ΔITD| (μs)	|ΔILD| (dB)	held-out SI-SDRi
Unprocessed	–	0.00	0.577	73.8	1.21	0.00
AsymDF	2	2.76	0.589	52.1	0.56	0.52
AsymDF	4	2.54	0.584	59.9	0.60	0.44
AsymDF (common head)	4	2.24	0.577	63.4	0.70	1.58
AsymDF (12k steps)	4	3.57	0.620	47.8	0.36	1.00
AsymDF (non-causal)	4	2.55	0.584	63.8	0.58	0.53
AsymDF (non-causal) (12k steps)	4	3.60	0.618	47.0	0.36	0.93
AsymDF	8	2.15	0.578	66.2	0.69	0.49
AsymDF	32	1.09	0.571	75.1	0.94	-0.61
AsymDF (12k steps)	32	1.98	0.575	69.3	0.73	0.31
AsymDF (non-causal)	32	1.23	0.570	73.2	0.91	0.05
Sym. window DF	2	2.73	0.592	57.3	0.55	0.93
Sym. window DF	4	2.52	0.586	63.4	0.60	0.79
Sym. window DF	8	1.99	0.579	70.0	0.74	0.55
Conv-TasNet	2	2.42	0.563	64.7	0.57	1.00
Conv-TasNet	4	2.10	0.554	68.8	0.64	1.10
Conv-TasNet (common head)	4	2.09	0.555	63.3	0.67	1.07
Conv-TasNet (non-causal)	4	2.20	0.549	61.6	0.59	1.05
Conv-TasNet	8	1.98	0.550	68.3	0.60	1.12
Conv-TasNet	32	1.71	0.548	71.2	0.61	0.46
Conv-TasNet (non-causal)	32	1.96	0.547	60.1	0.50	0.64

Table 1: Test-set results for every cell (2k steps unless noted). |ΔITD| and |ΔILD| are target-cue errors against the clean early target. The last column is SI-SDRi on the held-out set (CIPIC head, unseen noise environments and rooms).

met, but only because short hops train faster per step, so we treat it as uninformative rather than as confirmation. The cue criterion fails on timing: no model, and not even the oracle mask, keeps |ΔITD| under 20 μs, while |ΔILD| is comfortably under 1 dB. The window-versus-lookahead criterion does not falsify the hypothesis: lookahead buys nothing at a matched hop.

What this means for a hearing device. Within a 5 ms budget, the cheapest way to improve quality is not latency but capacity, data, and training. Strict causality removes almost nothing that a small model could use. A 4 ms AsymDF already streams at RTF 0.21 on one CPU thread. The open problem is binaural timing: a mask that leaves even a little noise in the low band moves the perceived location. That argues for cue-preserving structure (common gains below 1.5 kHz) or an ITD-specific

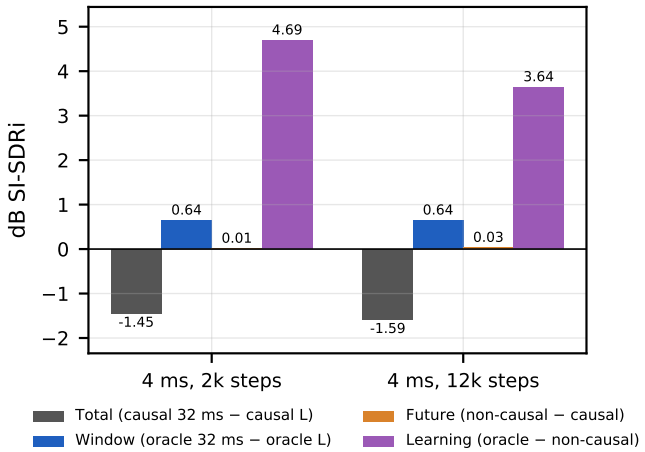


Figure 2: Where the 4 ms causal AsymDF stands relative to 32 ms, at two training budgets. The window cost is an oracle quantity; the future and learning terms come from the trained non-causal twin.

loss, not for more lookahead.

Limitations.

- The models are small (730k parameters for AsymDF) and far from the oracle, so whether the future becomes useful once the learning gap closes is untested. The GPU configuration (1–2M parameters, 40k steps, ≈100 h of speech) is sized to answer that.
- Training at equal steps confounds latency with supervision density. A frame-matched budget would be cleaner.
- The target keeps early reflections, so SI-SDRi is not comparable to dry-target benchmarks.
- There is one seed per cell.
- PESQ was unavailable on the build machine.
- All mixtures are simulated, although with real speech, real noise and measured heads.

6 Conclusion

For causal binaural enhancement inside a hearing-aid budget, we split the cost of low latency into three parts:

- the window, about 1 dB from 32 ms to 1 ms even for an ideal mask, and partly recoverable with asymmetric windows;
- the future, ≈0 dB for the networks we trained;
- learning, several dB, by far the largest.

Level cues survive and timing cues do not. A strictly causal 4 ms model runs comfortably in real time on one CPU thread. The practical message is that "hearing the future" is not what separates today's small causal models from good quality, and that the binaural ITD, not the SI-SDR, is the harder target. Code, configurations and the exact commands are in the project repository; `make all` reproduces every number in this paper.

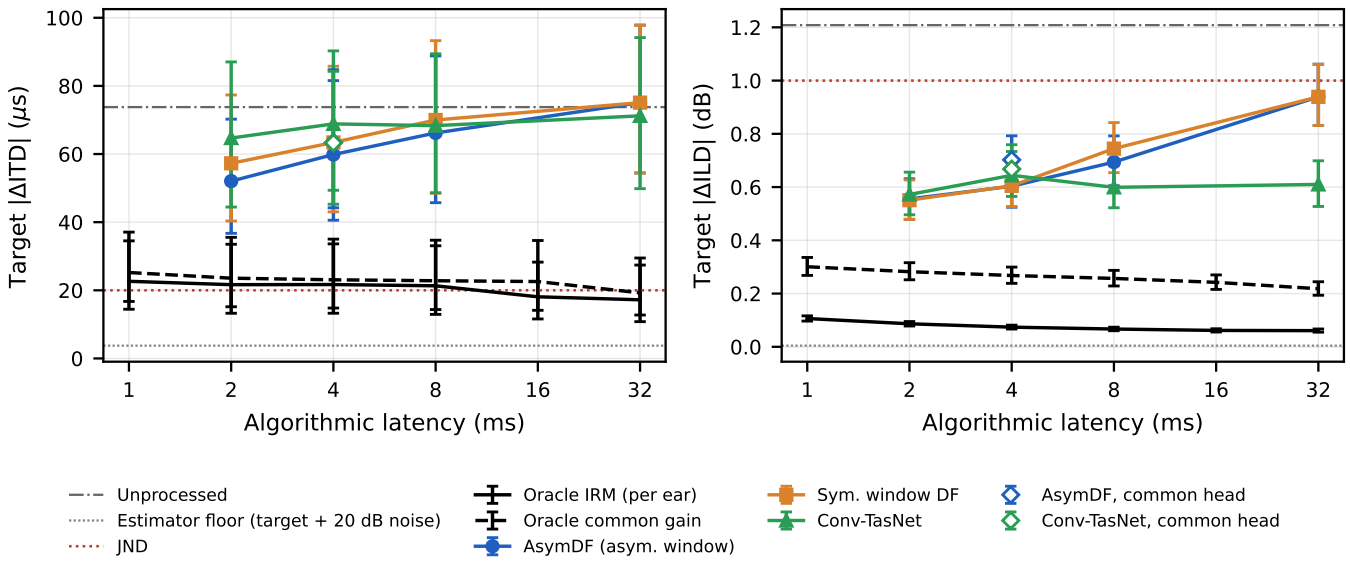


Figure 3: Target interaural-cue errors against the clean binaural target. Red: JND (20 μs, 1 dB). Grey dotted: the estimator's own floor (target plus independent noise at 20 dB SNR).

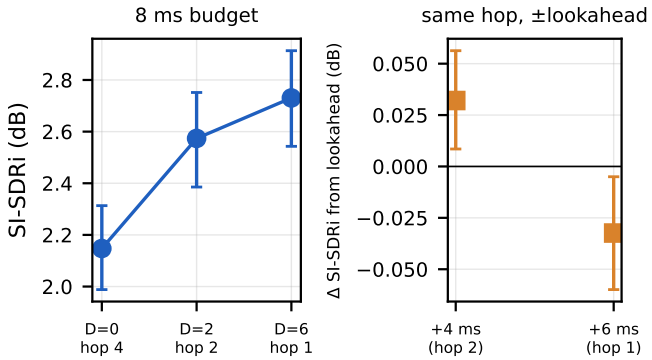


Figure 4: Left: an 8 ms budget split between synthesis window and lookahead ($L = (2{+}D)R$). Right: the same hop with and without D lookahead frames (paired per-utterance difference, 95% CI). The left-hand trend is the hop-size training effect, not the future.

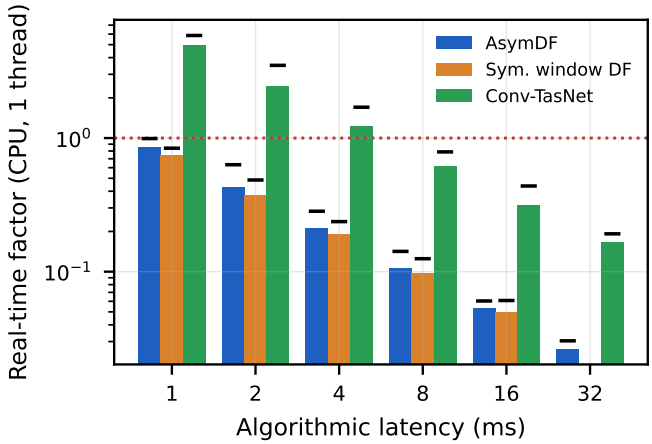


Figure 5: Streaming real-time factor on one CPU thread at batch size one (bars: median; ticks: p99). Below the red line the model keeps up.

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